

Quiz — 1.3.4 Web Technologies

Name: _____ Date: _____ Mark: _____ / 25

OCR H446 · Component 01 · 1.3.4

Section A — Multiple choice (8 marks)

1. The role of HTML in a web page is to define the: A. Colour and font styling B. Structure and content C. Server-side database logic D. Encryption of the page

Your answer: ☐

2. Which technology is primarily responsible for the presentation/styling of a page? A. HTML B. CSS C. JavaScript D. SQL

Your answer: ☐

3. A CSS selector beginning with # targets an element by its: A. tag name B. class C. id D. attribute value

Your answer: ☐

4. JavaScript is most commonly used to provide: A. Page structure B. Client-side interactivity and behaviour C. Database storage D. IP addressing

Your answer: ☐

5. In PageRank, a link from one page to another is treated as: A. A penalty B. A "vote" for the linked page's importance C. A keyword in the index D. An encryption key

Your answer: ☐

6. Links from high-ranking pages in PageRank are: A. Ignored B. Worth more than links from low-ranking pages C. Always worth the same as any other link D. Counted as negative votes

Your answer: ☐

7. Which is an example of server-side processing? A. Instant form validation in the browser B. Animating a button on hover C. Checking login credentials against a database D. Showing/hiding a menu on click

Your answer: ☐

8. Why must security-critical validation also be performed server-side? A. It runs faster on the client B. Client-side checks can be viewed and bypassed by the user C. The server cannot run JavaScript D. It reduces server load

Your answer: ☐

Section B — Short answer (12 marks)

9. State the role of each of HTML, CSS and JavaScript in a web page. [3]

10. Describe how a search engine builds its index of web pages. [2]

11. Explain the purpose of the damping factor in the PageRank algorithm. [2]

12. PageRank is calculated iteratively. Explain what this means and when it stops. [2]

13. Give one advantage and one disadvantage of using client-side processing instead of server-side processing. [2]

14. State one task that is better suited to server-side processing and explain why. **[1]**

Section C — Exam-style question (5 marks)

Explain why the email check can be done client-side but the payment verification must be done server-side, referring to speed, server load and security. [5]

[illegible]